Metadata

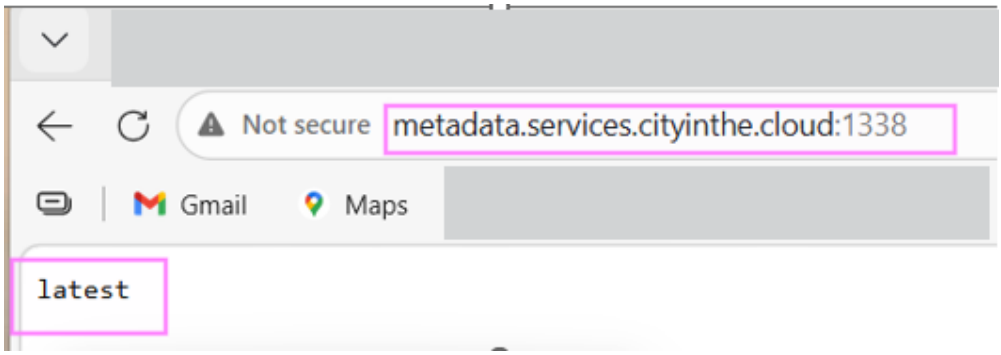
Scenario Overview

During routine monitoring, the security team discovered a server exposing cloud instance metadata over an accessible HTTP endpoint. Metadata services often reveal critical information about cloud infrastructure, including availability zones, IAM roles, instance types, and network details. Your task is to explore the metadata service, enumerate the available endpoints, and extract key configuration details. This includes identifying the instance's availability zone, IAM role, instance type, operating system, and ultimately recovering the hidden flag.
Tasks

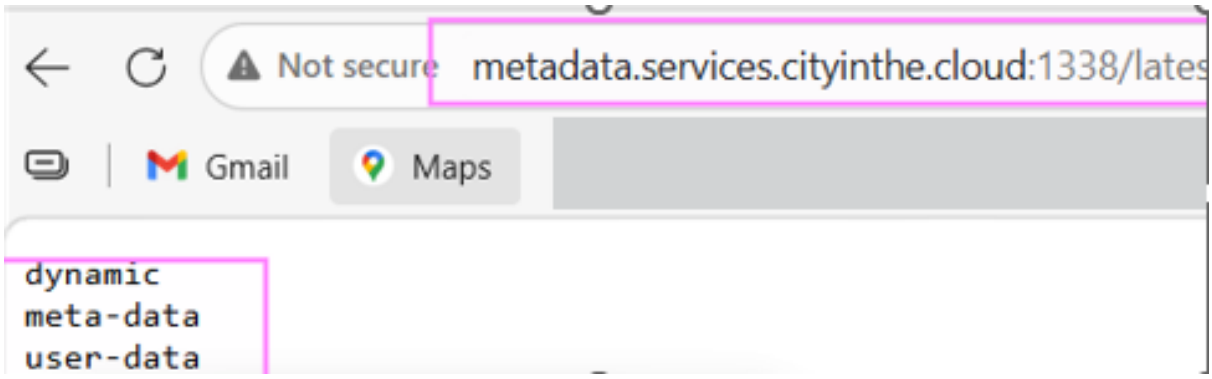
1. Verify access to the metadata service
    - Navigate to the base metadata endpoint.
    - Confirm that the server responds with the expected directory structure (e.g., /latest).
2. Identify the availability zone
    - Explore the meta-data directory.
    - Locate the placement subdirectory.
    - Retrieve the value of the availability-zone endpoint.
3. Determine the IAM security credentials role
    - Navigate to the iam/security-credentials directory.
    - Identify the name of the role assigned to the instance.
4. Identify the instance type
    - Query the instance-type endpoint within the metadata service.
    - Record the instance type returned.
5. Determine the operating system name and version
    - Retrieve the AMI ID from the metadata service.
    - Use an AMI lookup tool (such as the Ubuntu Amazon EC2 AMI Finder) to identify the OS name and version associated with the AMI.
6. Recover the flag
    - Enumerate the network interface metadata.
    - Identify the MAC address and use it to explore sub-endpoints.
    - Locate the endpoint containing the hidden flag and document it.

1. **What availability zone is this instance hosted in?** (Access the placement/availability-zone endpoint)

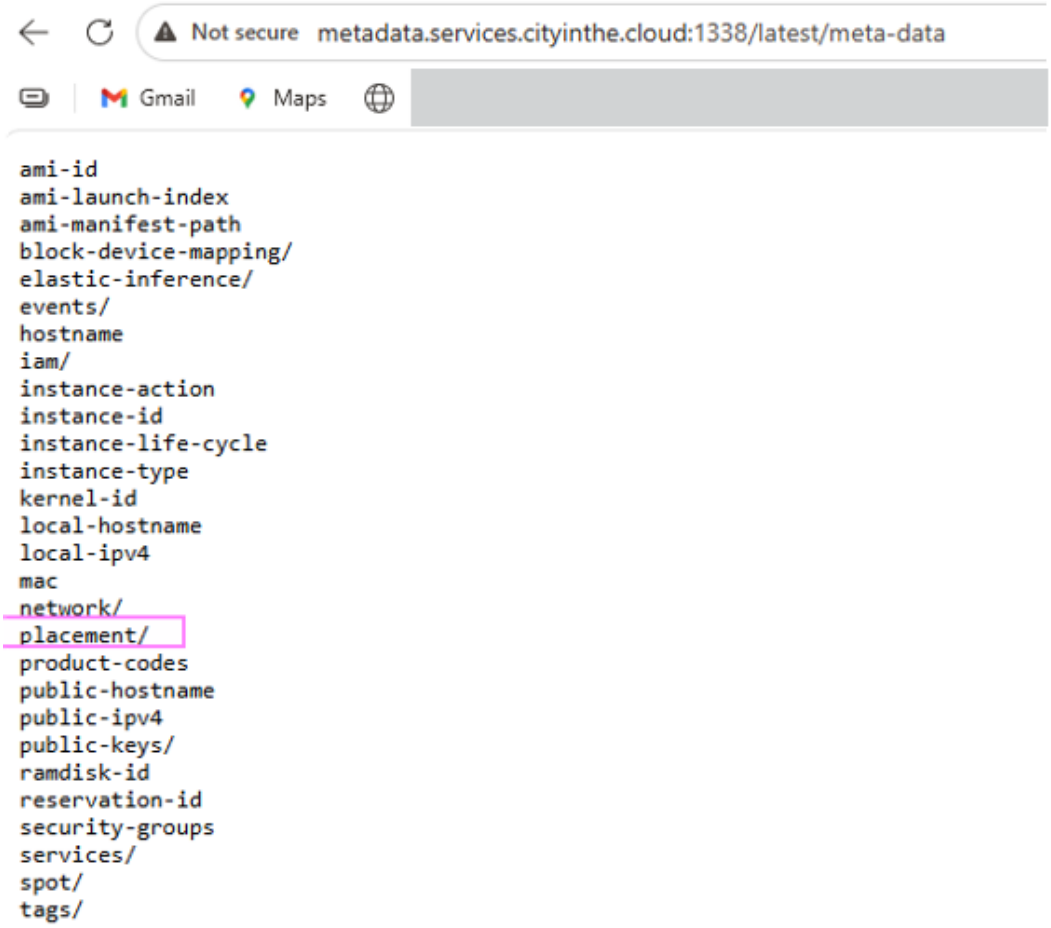
Copy/paste the URL and see the output ‘’latest,” which confirms it is a http web server.



At the end of the URL add **/latest** You will then see a list.
http://metadata.services.cityinthe.cloud:1338/latest

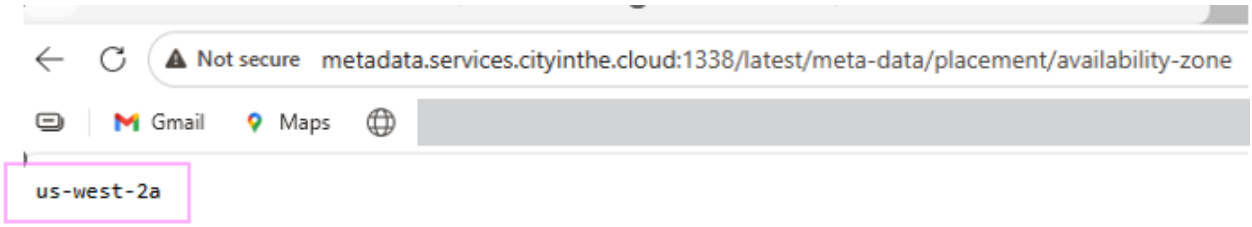


Access the meta-data endpoint by adding **/meta-data** to the URL: http://metadata.services.cityinthe.cloud:1338/latest/meta-data You will then see "placement."



Add at the end of the URL **/placement/availability-zone**: http://metadata.services.cityinthe.cloud:1338/latest/meta-data/placement/availability-zone

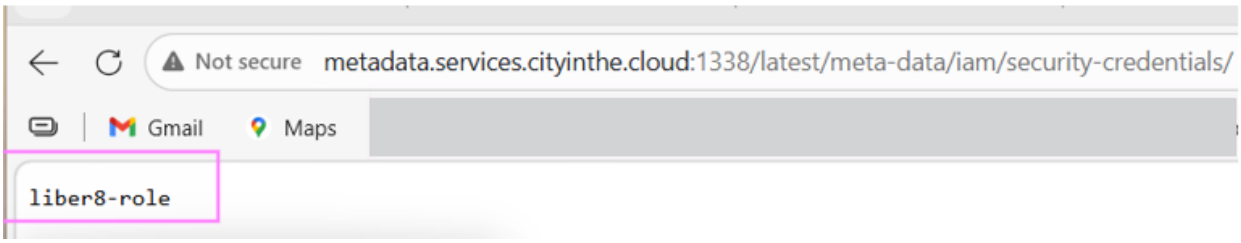
Answer: us-west-2a



2. **What is the security credential's role named?** (Access the iam/security-credentials endpoint)

   Answer: liber8-role

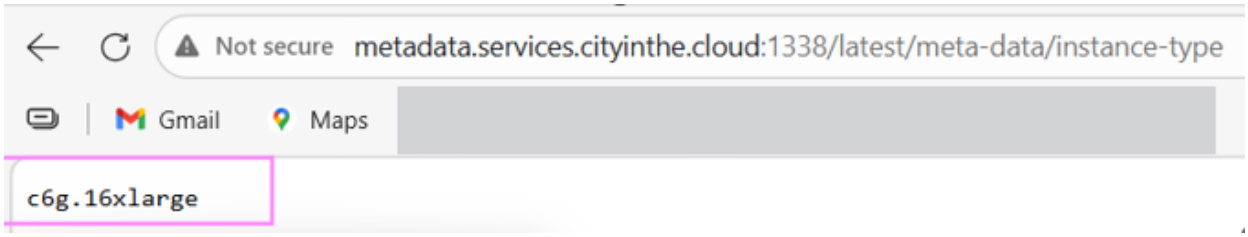
Add at the end of the URL "metadata.services.cityinthe.cloud:1338/latest/meta-data"add **iam/security-credentials (**metadata.services.cityinthe.cloud:1338/latest/meta-data/iam/security-credentials/**)**



3. **What is the instance type being used?** (Access the instance-type endpoint)

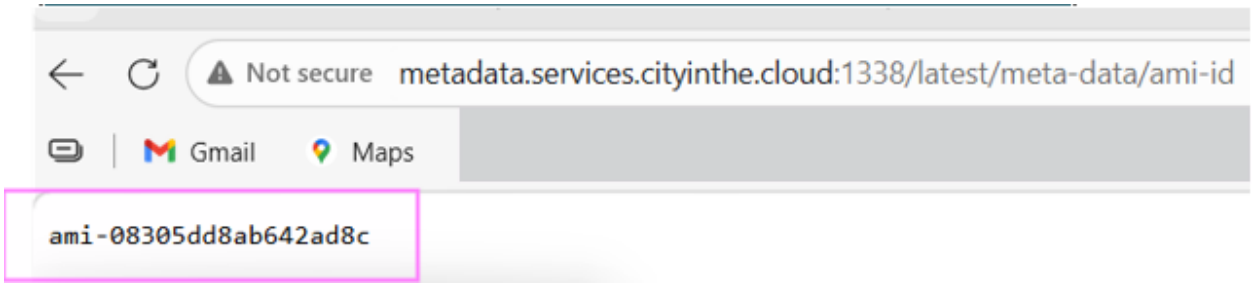
Answer: c6g.16xlarge

At the end of the URL "metadata.services.cityinthe.cloud:1338/latest/meta-data"add **/instance-type** (**http://metadata.services.cityinthe.cloud:1338/latest/meta-data/instance-type**)

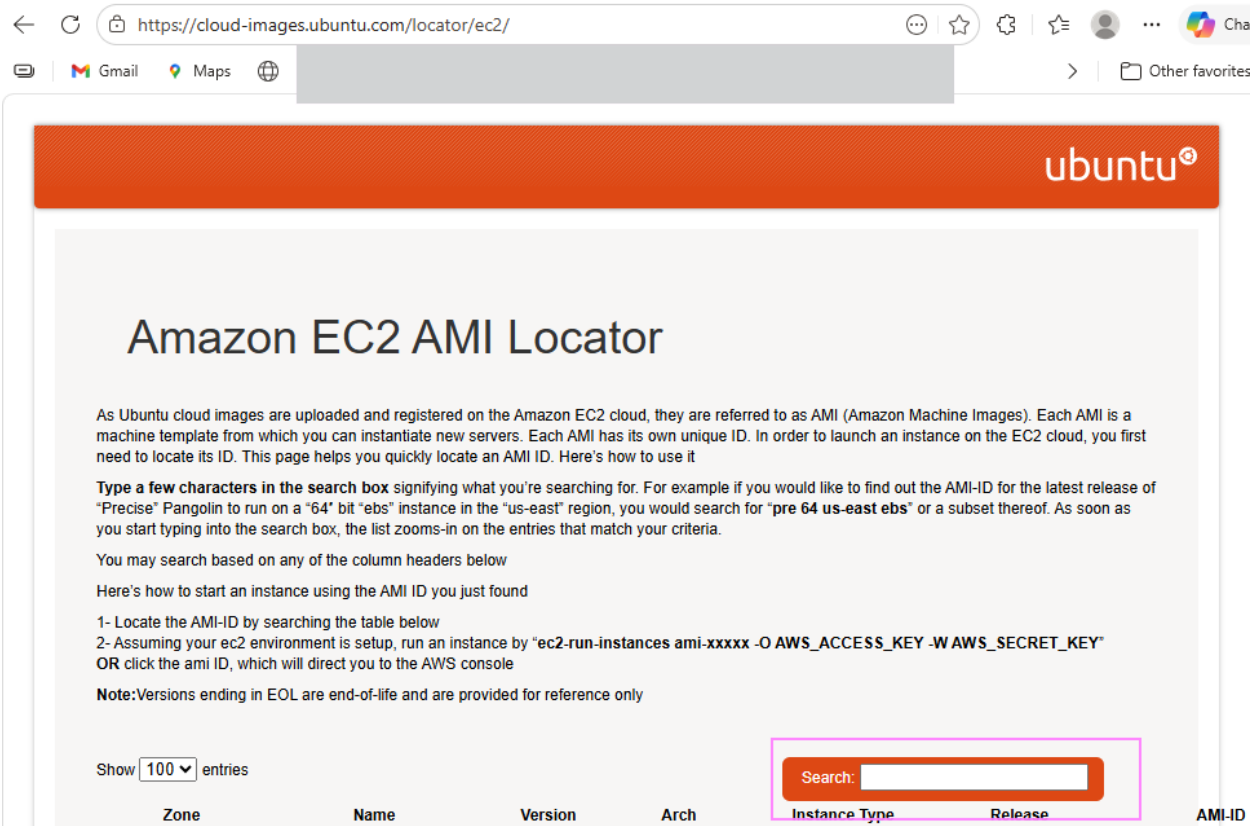


4. **What is the operating system name and version number?** (Access the ami-id endpoint first)

At the end of the URL "metadata.services.cityinthe.cloud:1338/latest/meta-data"add **/ami-id** (**http://metadata.services.cityinthe.cloud:1338/latest/meta-data/ami-id**)



Next, copy/paste the AMI ID in the search box using Ubuntu Amazon EC2 AMI Finder



You will then see the name and version

Answer: Xenial Xerus16.04 LTS

Amazon EC2 AMI Locator

As Ubuntu cloud images are uploaded and registered on the Amazon EC2 cloud, they are referred to as AMI (Amazon Machine Images). Each AMI is a machine template from which you can instantiate new servers. Each AMI has its own unique ID. order to launch an instance on the EC2 cloud, you first need to locate its ID. This page helps you quickly locate an AMI ID. Here's how to use it

Type a few characters in the search box signifying what you're searching for. For example if you would like to find out the AMI-ID for the latest release of "Precise" Pangolin to run on a "64" bit "ebs" instance in the "us-east" region, you would searc "**pre 64 us-east ebs**" or a subset thereof. As soon as you start typing into the search box, the list zooms-in on the entries that match your criteria.

You may search based on any of the column headers below

Here's how to start an instance using the AMI ID you just found

1- Locate the AMI-ID by searching the table below
2- Assuming your ec2 environment is setup, run an instance by "**ec2-run-instances ami-xxxxx -O AWS_ACCESS_KEY -W AWS_SECRET_KEY**"
OR click the ami ID, which will direct you to the AWS console

Note: Versions ending in EOL are end-of-life and are provided for reference only

Show 100 entries

Search: ami-08305dd8ab642ad8c

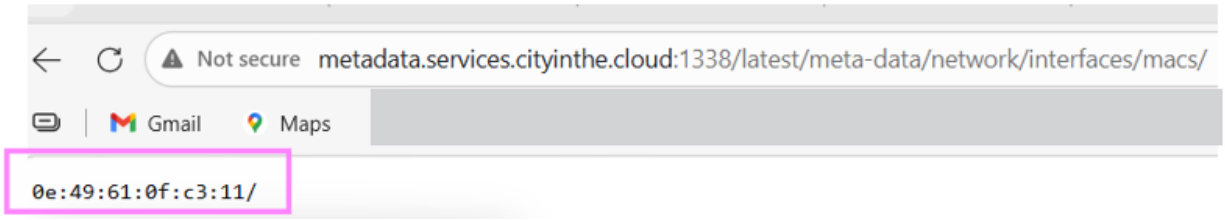
Zone	Name	Version	Arch	Instance Type	Release	AMI-ID	AKI-ID
us-west-2	Xenial Xerus	16.04 LTS	ar 64	hvm:ebs-ssd	20210928	ami-08305dd8ab642ad8c	hvm
Any	Any	Any	Any	Any	Any	Any	Any

Showing 1 to 1 of 1 entries (filtered from 500 total entries)

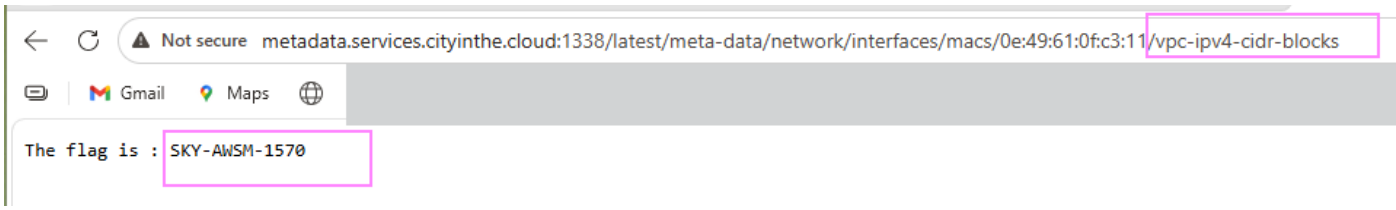
5. What is the flag?

Answer: SKY-AWSM-1570

Search for the MAC Address: http://metadata.services.cityinthe.cloud:1338/latest/meta-data/network/interfaces/macs/



Next, replace **<MAC>** http://metadata.services.cityinthe.cloud:1338/latest/meta-data/network/interfaces/macs/**<MAC>**/ with the MAC address 0e:49:61:0f:c3:11/ to search for sub-endpoints.

Next, I the sub-endpoints at the end of the URL: metadata.services.cityinthe.cloud:1338/latest/meta-data/network/interfaces/macs/0e:49:61:0f:c3:11/vpc-ipv4-cidr-blocks

Answer: SKY-AWSM-1570